

Analysis of the Impact of Feature Map Resolution on Object Localization Performance:

A Comparative Study of ResNet-50-based CAM and Grad-CAM

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Abstract

This study investigates the impact of feature map resolution on object localization precision by conducting comparative experiments between CAM and Grad-CAM using ResNet-50 pre-trained on the ImageNet dataset[4]. For the analysis, feature maps were extracted from Layer 4 (7×7) and Layer 3 (14×14) of the bottleneck structure, and Intersection over Union (IoU) performance was measured across various scenarios using the Stanford Dogs dataset[3]. Experimental results demonstrate that while Layer 3-based Grad-CAM, which provides higher-resolution feature maps, excels in tiny object detection, CAM exhibits superior robustness for partially occluded objects. These findings suggest that selecting the appropriate layer based on the visualization objective is essential, regardless of the model's architectural depth.

1. Introduction

Explainable Artificial Intelligence (XAI) is a pivotal technology that provides reliability by visualizing the decision-making rationale of black-box models. For deep and complex architectures such as ResNet-50[4], the precision of these explanations varies significantly depending on the specific stage at which features are visualized. This study aims to quantitatively analyze the impact of resolution disparities arising from the bottleneck layer structures of ResNet-50 on the precision of object localization.

2. Background and Related Work

2.1. CAM (Class Activation Map): Visualization Principle Based on Global Average Pooling

CAM is a technique used to visualize the specific regions of an image that a CNN model focuses on when identifying a particular class[1]. The fundamental mechanism involves replacing the traditional Fully Connected Layer with

a Global Average Pooling (GAP) layer immediately following the final convolutional layer. The final activation map is generated by computing the spatial average of each feature map and then performing a weighted sum using the class-specific weights from the output layer. Although this approach effectively highlights class-wise contributions while preserving spatial localization, it imposes an architectural constraint, as the model must specifically incorporate a GAP layer.

2.2. Grad-CAM: A General Visualization Technique Using Gradient Information

Grad-CAM was proposed to overcome the structural constraints of CAM by utilizing gradient information generated during the backpropagation process to calculate weights[2]. It computes the importance of each feature map by calculating the gradient of a specific class score with respect to the feature maps and then combines this information with the feature maps to construct a visualization map. Subsequently, a ReLU operation is performed to extract only the regions with a positive influence, thereby identifying meaningful areas. Grad-CAM is applicable to all CNN-based architectures regardless of the presence of a GAP layer and provides the flexibility to visualize any specific layer of choice.

2.3. Relationship Between Feature Map Resolution and Receptive Field

In CNN architectures, as the layers deepen, the receptive field covered by the model expands, while the resolution of the feature maps gradually decreases during the computational process. In the case of the ResNet-50 architecture[4] used in this study, the final layer, Layer 4, has a low resolution of 7×7 . While this is advantageous for identifying the global context of an image for class classification, it possesses limitations in precisely localizing small objects due to the coarse granularity of the pixel units. In contrast, the preceding stage, Layer 3, maintains a relatively higher reso-

lution of 14×14 , enabling more granular localization. This study focuses on analyzing the impact of these resolution disparities on visualization quality, specifically in relation to object size and environmental noise.

3. Research Methodology and Structure

3.1. Experimental Environment and Model Configuration

All experiments in this study were conducted using the open-source deep learning framework PyTorch[5]. The ResNet-50 architecture, widely recognized for its performance in image classification and feature extraction, was adopted as the baseline model. To evaluate general visualization performance, we utilized weights pre-trained on the ImageNet dataset[4]. The experimental environment was configured to collect internal feature maps and gradient information generated during backpropagation, enabling the generation of both CAM and Grad-CAM visualizations.

3.2. Dataset and Preprocessing

The Stanford Dogs Dataset[3], which contains images of 120 different dog breeds, was employed for the experiments. To optimize the data for the model’s input specifications, all images were resized to 224×224 pixels. Normalization was performed using the RGB channel-wise mean and standard deviation consistent with the pre-trained ImageNet model’s distribution. To quantitatively measure the localization accuracy of the visualization maps, ground truth bounding box coordinates were extracted from the XML metadata of each image and set as the evaluation baseline.

3.3. Evaluation Metric: IoU (Intersection over Union)

The Intersection over Union (IoU) metric was used to quantitatively assess the localization performance of the generated heatmaps. IoU represents the degree of overlap between the predicted object region and the actual ground truth, calculated as the ratio of the area of intersection to the area of union between the two regions. In this study, a predicted box was defined by thresholding the heatmap based on the maximum activation point, and the degree of overlap with the ground truth box was calculated; values closer to 1 signify more precise localization.

3.4. Experimental Design and Resolution Comparison

To analyze the impact of feature map resolution on localization performance, we designed a comparative experiment targeting different layers of ResNet-50. The first configuration utilizes CAM based on the feature maps of Layer 4, the model’s final convolutional layer, which has a low resolution of 7×7 . The second configuration utilizes Grad-CAM

based on the feature maps of Layer 3, an earlier stage that maintains a resolution of 14×14 —four times higher than that of Layer 4. This study applies these two resolution settings across four scenarios (single object, multiple objects, tiny objects, and partial occlusion) to conduct an in-depth analysis of visualization precision changes relative to object size and environmental noise.

4. Experimental Results and Discussion

4.1. Detailed Experimental Analysis by Scenario

In this section, we conduct a comparative analysis of the impact of various environmental constraints on object localization performance using the Intersection over Union (IoU) metric. To ensure the reliability of the findings, the data from the baseline (Original) environment are contrasted with the detailed analysis results for each specific scenario. The following results present the performance analysis based on object and environmental characteristics, derived from a synthesis of quantitative metrics and qualitative visualization data.

Severe Distortion: In this scenario, we analyzed the robustness of localization performance based on feature map depth under conditions where severe distortion was applied to the entire image.

As shown in Table 1, the high-resolution Layer 3-based method (0.4034) yielded lower performance compared to the Layer 4-based method (0.5367), despite the latter having a lower spatial resolution. This disparity stems from the difference in the nature of the information extracted by each layer.

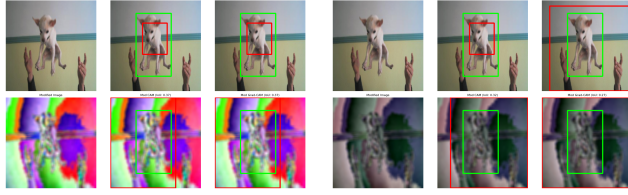
Table 1. Comparison of IoU Scores by Layer under Severe Distortion

Method (Layer)	IoU Score
CAM / Grad-CAM (Layer 4)	0.5367
Grad-CAM (Layer 3)	0.4034

The performance gap can be attributed to the distinct types of features prioritized at different depths. Layer 3 tends to focus on fine Local Features, such as the eyes, nose, and mouth of the object. When the entire image is distorted, these subtle feature signals are easily overwhelmed or destroyed by noise, resulting in a sharp decline in localization precision.

In contrast, Layer 4 maintains the Global Context, encompassing the overall shape of the object and its relationship with the surrounding environment, even at a lower resolution. Consequently, it demonstrates superior stability in preserving positional information without losing the general form of the object, even under severe environmental noise.

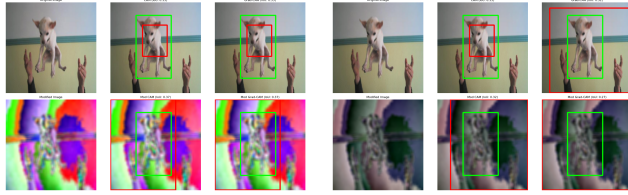
Tiny Object Detection (Tiny Dog): In scenarios involving



(a) Layer 4-based Visualization

(b) Layer 3-based Visualization

Figure 1. Comparison of visualization results in the Severe Distortion scenario. (a) maintains the global context and encompasses the object region despite the lower resolution, whereas (b) shows the destruction of local feature points due to noise.



(a) CAMvsGrad-CAM (Layer 4)

(b) CAMvsGrad-CAM (Layer 3)

Figure 2. Comparison of visualization results in the Severe Distortion scenario. (a) maintains the global context despite the lower resolution, whereas (b) shows the destruction of feature points caused by noise.

extremely small objects, feature map resolution was identified as a critical factor determining localization performance.

As shown in Table 2, when utilizing the same Layer 4, both CAM and Grad-CAM recorded a very low IoU of 0.0102, demonstrating that both methods share the same inherent limitations at this depth.

Table 2. Tiny Dog: IoU Performance Comparison (Layer 4)

Method	IoU Score
CAM (Layer 4)	0.0102
Grad-CAM (Layer 4)	0.0102

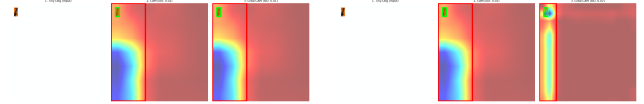
In contrast, comparative experiments across layer depths (Table 3) revealed that Layer 3-based Grad-CAM (0.0207), with its higher resolution, exhibited a significant performance advantage—more than double that of Layer 4-based CAM (0.0102).

Table 3. Tiny Dog: IoU Performance Comparison by Resolution

Method (Layer)	IoU Score
CAM (Layer 4, 7×7)	0.0102
Grad-CAM (Layer 3, 14×14)	0.0207

The basis for this performance disparity lies in visualization precision. While the final layer, Layer 4 (7×7),

possesses a broad receptive field advantageous for class classification, its pixel units are too coarse for representing minute objects. This leads to activation maps that bleed significantly beyond the object’s boundaries. Conversely, the preceding stage, Layer 3 (14×14), provides four times the spatial resolution, enabling the model to precisely pinpoint tiny objects even when they are located in the corners of an image. These results suggest that utilizing high-resolution feature maps is essential for ensuring localization precision (IoU) as object size decreases.



(a) CAM vs. Grad-CAM (Layer 4)

(b) CAM vs. Grad-CAM (Layer 3)

Figure 3. Comparison of visualization results in the Tiny Dog scenario. (a) shows blurred activation regions due to low resolution, whereas (b) demonstrates sharp and precise localization facilitated by higher spatial resolution.

Partial Occlusion: In scenarios where the object was partially obscured by other elements, CAM (Layer 4) demonstrated significantly superior performance compared to Grad-CAM (Layer 3).

First, as illustrated in Table 4, both methods recorded an identical IoU of **0.4780** when based on the same Layer 4, further substantiating their mathematical equivalence at the final convolutional layer.

Table 4. Partial Occlusion: IoU Performance Comparison (Layer 4)

Method	IoU Score
CAM (Layer 4)	0.4780
Grad-CAM (Layer 4)	0.4780

In contrast, the comparative results across layer resolutions in Table 5 reveal that while CAM (Layer 4) maintained a score of 0.4780, Grad-CAM (Layer 3) dropped sharply to 0.2277, representing a decline of more than 50%.

Table 5. Partial Occlusion: IoU Performance Comparison by Layer Resolution

Method (Layer)	IoU Score
CAM (Layer 4)	0.4780
Grad-CAM (Layer 3)	0.2277

These findings can be attributed to the computational characteristics of the Intersection over Union (IoU) metric and the distinct activation tendencies of each layer. The high-resolution Layer 3 tends to react with high sensitivity only to specific exposed feature points (e.g., ears, legs). This

leads to an overly narrow predicted bounding box that fails to encompass the overall geometry of the object, thereby reducing the intersection area with the ground truth and degrading IoU performance.

Conversely, the final Layer 4 possesses a broader and coarser activation range. This allows the model to contextually identify the entire "object region" as a single cohesive activation cluster, even when parts of the object are obscured. Consequently, the predicted bounding box encompasses the object more comprehensively, maximizing the overlap with the ground truth and ensuring stable localization performance.

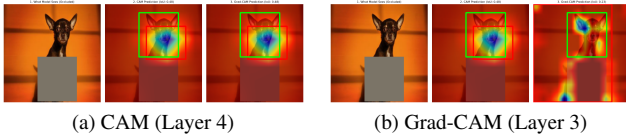


Figure 4. Comparison of visualization results in the Partial Occlusion scenario. (a) shows broad activation that encompasses the occluded regions, whereas (b) demonstrates localized activation limited to specific exposed feature points.

5. Conclusion

This study quantitatively analyzed the impact of feature map resolution on the quality of visualization-based object localization within the ResNet-50 architecture. The key conclusions derived from comparative experiments across various environmental constraint scenarios are as follows.

First, when sharing the same final layer (Layer 4), CAM and Grad-CAM produce mathematically identical activation weights, resulting in consistent quantitative IoU performance. This empirically validates the mathematical equivalence of both techniques in models adopting a Global Average Pooling (GAP) structure.

Second, a clear trade-off exists between feature map resolution and the level of information abstraction. While high-resolution visualization based on Layer 3 demonstrated overwhelming precision in detecting tiny objects (Tiny Dog), Layer 4-based visualization—which preserves global context—exhibited higher robustness in complex environments characterized by severe image distortion or partial occlusion.

Third, the results demonstrate that IoU performance does not necessarily improve with higher resolution alone; rather, the optimal layer selection must be adapted based on object size and environmental noise characteristics.

Figure 5 illustrates the overall workflow of the experiments conducted in this study and the optimal visualization strategies for each environmental condition. The findings suggest that an "Adaptive Layer Selection Strategy," which accounts for environmental data constraints, is essen-

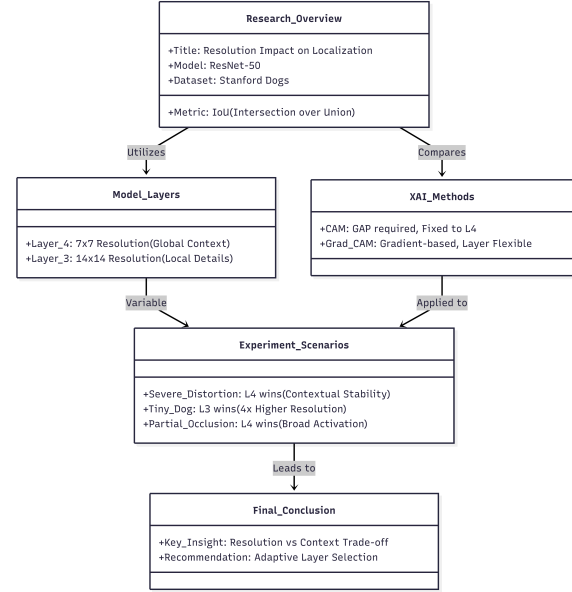


Figure 5. Comprehensive summary image illustrating the overall experimental design and the derived optimal layer selection strategies for each scenario.

tial when applying XAI technologies to practical real-world domains.

References

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